

# VERONICA DADZIE MANSO

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## EDUCATION

**TU Munich, TU Wien, TU Dresden, and the University of Twente**

**October 2025 - Present**

Erasmus Mundus Joint Master's in Cartography - European Union Scholar

**University of Mines and Technology (UMaT), Ghana**

**September 2019 - September 2023**

BSc Geomatic Engineering

**First Class Honours (81.88 /100, GPA 3.8/4)**

## RESEARCH INTERESTS

Geo-Health, Cartography, Spatial Analysis, Remote Sensing and GIS Applications

## RESEARCH EXPERIENCE

**Evaluating OCHA Humanitarian Icons under a Natural Disaster Frame: Interpretation, Perceived Icon Characteristics, and User Priorities**  
**May 2026 - July 2026**

*Manuscript in Preparation for Peer-Reviewed Submission (Co-author)*

- Analyzed icon-interpretation responses from 70 participants, providing the study's accuracy data, and authored the Participants section of the manuscript
- Shelter, prioritized by 88.6% of participants, was correctly recognized by only 57%, revealing a gap between icon importance and recognizability
- Perceived semantic distance was identified as the strongest predictor of correct interpretation across all 15 icons tested, exceeding the effects of visual simplicity and user expertise

**Spatial Hotspots of Mpox Risks in Democratic Republic of the Congo: A Health-Zone Analysis**

**October 2025 - February 2026**

*Geo-Health Project (Supervisor: Prof Justine Blanford and Prof Yanchao Cheng)*

- Applied five spatial-weighting methods (Queen contiguity, fixed distance band, k-nearest neighbors) to map Mpox risk across 200 health zones in the DR Congo
- Found significant positive spatial autocorrelation ( $p < 0.001$ ) across all five methods, with  $k = 4$  nearest neighbors ( $I = 0.259$ ,  $z = 5.86$ ) yielding the highest result
- Findings supported prioritizing surveillance and diagnostic resources in Sankuru, Tshuapa, Sud-Kivu, and Maniema provinces, based on persistent hotspots identified across all methods

**Complex-valued Convolutional Neural Network for Land-Cover Classification**

**October 2025 - February 2026**

*Big Geospatial Data Project (Supervisor: Dr Kwasi Nyako Poku-Agyemang)*

- Contributed to automated land-cover classification of agricultural land into 15 classes using SAR data, comparing Complex-valued CNN (CV - CNN) and Real-valued CNN (RV - CNN) approaches
- CV-CNN attained 93.21% accuracy compared to 92.14% for RV - CNN (Cohen's  $\kappa = 0.942$  vs. 0.928), with the largest gain observed for crops such as rapeseed (+18.83% in F1-score)

**Data Quality Assessment of Geodetic Observations Using Statistical Testing Procedures**

**January 2023 - August 2023**

*Geomatic Engineering undergraduate thesis (Supervisor: Prof Yao Yevenyo Ziggah)*

- Performed data validation (outlier detection) on geodetic observations using threshold values of robust estimation methods such as Danish, Thompson Tau and RANSAC using MATLAB.
- All the robust estimation methods can detect and handle outliers in the observations with satisfactory accuracy levels. Nevertheless, Random Sample Consensus (RANSAC) algorithm conceded the greatest number of outliers.

**Analysis of Land Use / Land Cover Change for Tarkwa-Nsuaem District - Ghana**

**March 2023 - April 2023)**

*Urban Planning Project for Tarkwa-Nsuaem District (Supervisor: Dr. Micheal Aduah)*

- Analyzed Sentinel-2 imagery (2016 - 2021) into 4 land cover classes using Google Earth Engine and ArcGIS Pro
- Settlement expanded from 50.240% to 53.518%, while Vegetation declined from 33.687% to 33.120%, Bareland from 15.962% to 13.276%, and Water Bodies from 0.112% to 0.086%
- The research supported zoning enforcement and green-space protection of the Tarkwa-Nsuaem District

## TEACHING AND WORK EXPERIENCE

### Teaching and Research Assistant UMaT Geomatic Department, Ghana

**November 2023 - September 2025**

- Supervised first- and second-year students' field and research work in Urban Planning and Introduction to Geomatics.
- Grouped students to work on projects in Cartography, Remote Sensing and GIS Applications, and present their findings, building their ability to communicate geospatial analysis
- Organized tutorial sessions for first year undergraduate students on QGIS and ArcGIS software, strengthening their ability to apply geospatial concepts to real-world problems
- Supported my supervisor with research (GNSS receiver operation, data collection, and surveying)
- Marked and graded students' assignments and exam scripts for Cartography, Remote Sensing and GIS Applications

### Intern, Achiever's Geo-Consult, Ghana

**October 2022 - January 2023**

- Computed stockpile volumes using Geovia Surpac to support accurate resource estimation for mine planning
- Set out and demarcated lands using Total Station and GNSS/RTK receivers, to establish precise boundary points for parcel allocation and registration
- Collaborated with other surveyors to survey a community displaced by mining operations and mapped affected properties to support compensation and planning of the resettlement of the community.

### Intern, Geo-Tech Surveys, Ringway Estate - Osu, Ghana

**May 2022 - August 2022**

- Prepared cadastral plans using AutoCAD for land registration for clients, solving land ownership issues in Ghana
- Collaborated with geologists to collect soil samples across flood-affected areas in Ghana to support assessment of soil infiltration characteristics for a proposed drainage system design to mitigate urban flooding

## PROFESSIONAL CERTIFICATE

- Transforming Maps into Stories and Data into Decisions - RCMRD, Kenya
- AI Fluency: Framework & Foundations - Anthropic, Coursera
- Neural Networks and Deep Learning - DeepLearning.AI, Coursera
- Data Analysis with R programming - Google, Coursera
- Spatial Analysis and Satellite Imagery in GIS - University of Toronto, Coursera
- Mastering Geospatial Analysis with QGIS - L&T EduTech, Coursera

**Ongoing**  
**June 2026**  
**September 2025**  
**April 2024**  
**March 2024**  
**January 2024**

## HONOURS AND AWARDS

- **Erasmus Mundus Scholarship Award**, European Commission **2025 - Present**
- **Best Poster Presentation Award**, Technical University of Munich **2025**
- **Best Female Graduating Student**, Geomatic Eng Department, UMaT **2023**
- **Beneficiary**, Total Ghana Scholarship **2021 - 2023**
- **Departmental Excellence Award, UMaT** **2022**
- **Beneficiary**, Ghana Chamber of Mines Tertiary Education Fund **2019 - 2021**

## PROFESSIONAL AFFILIATIONS / MEMBERSHIPS

- Next Generation Cartographers (NGC)
- International Society for Photogrammetry and Remote Sensing (ISPRS)
- Ghana Geospatial Society, Ghana (GGS)
- Ghana Institution of Surveyors (GhIS)
- Australasian Institute of Mining and Metallurgy (AusIMM), Ghana
- Women in Engineering (WINE), UMaT, Ghana

## SEMINARS AND CONFERENCES

- **International Federation of Surveyors (FIG)** - Your World, Our World: Resilient Environment and Sustainable Management for All. **May 2024**

- **FIG Climate Compass Task Force Seminar** - How to Use a Range of Survey Tools to Monitor Flooding from Tropical Cyclones **March 2024**
- **Ghana Geospatial Society Annual conference** - Harnessing Geospatial Technology for the Achievement of the SDGs in Ghana **November 2023**
- **Geovia User Conference** - Dassault Systemes upgrade on Surpac Software **July 2023**
- **Geomatic Engineering Student Association Seminar** - The Use of Drones in Geomatics **November 2022**
- **AusiMM Tarkwa Student Chapter Conference** - Bridging the Gap Between Academia and Industry - A Catalyst for a Skilled Workforce **September 2022**

## PRESENTATIONS

- **Poster Presentation, Technical University of Munich, Germany** **February 2026**  
Presented on: "Complex-valued Convolutional Neural Network for Land-Cover Classification"
- **Geomatic Engineering Department, Annual Synopsis Seminar** **July 2023**  
Presented on: "Data Quality Assessment of Geodetic Observations Using Statistical Testing Procedures"

## SOFTWARE / TOOLS

- **Spatial Statistics and Epidemiology:** GeoDa, SaTScan
- **GIS and Remote Sensing:** ArcGIS Pro, QGIS, Google Earth Engine
- **Programming:** R, MATLAB, JavaScript, Python
- **Cartography and Web Mapping:** Leaflet.js, MapLibre GL JS, Maputnik and Affinity Designer
- **Others:** Microsoft Office Suite, Geovia Surpac, AutoCAD

## OTHER SKILLS AND COMPETENCIES

- Excellent team player, self-motivated and results-oriented
- Critical analyst and very innovative
- Easily adaptable to changes and readiness to tackle new problems

## LEADERSHIP EXPERIENCE

- **Assistant Ambassador - Innovation Fair** **2023**
- **Class Representative** **2022 - 2023**
- **Ladies Wing Vice President:** UMaT **2022 - 2023**
- **Mentor** - Geomatic Engineering Student Association, UMaT **2022**
- **Main Women Commission Organizer:** UMaT **2021 - 2022**
- **Course Representative** - Geomatic Engineering Department **2021**

## LANGUAGE

- Good English communication skills (Distinguished levels in Listening, Speaking, Reading and Writing)
- French and German at Elementary level

## COMMUNITY / VOLUNTEERING SERVICES

- Eudora Care Foundation UMaT, Tarkwa** **September 2023 - Present**
- I support mentoring underprivileged high school students by providing career guidance and academic support
  - Donate food, sanitary pads, and stationery to underserved schools across the Western Region of Ghana

- International Federation of Surveyors (FIG) Working Week** **May 2024**

- Assisted with participant registration, welcoming attendees, and providing directions at the conference.
- Supported the setup and preparation of presentation venues and conference facilities.
- Assisted in the distribution of conference materials and souvenirs to participants.

- Students in Community Services** **September 2022**

- Provided clothing and healthcare to underserved communities in Ghana's Western and Central Regions.
- Conducted teaching and training exercises for the students in the communities

## REFERENCES

Available upon request